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Cyber Forensics and Malware

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Case Study Abstract

**Forensic Reconstruction of a Conti Ransomware
Incident Through Memory and Storage Access
Artifact Analysis**

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Forensic Reconstruction of a Conti Ransomware Incident Through Memory and Storage Access Artifact Analysis

Abstract

Ransomware attacks pose a significant threat to organizational systems by encrypting critical data and disrupting normal operations. This case study focuses on the forensic reconstruction of a suspected **Conti ransomware incident on a Windows-based corporate workstation**. The scenario assumes that an employee workstation exhibits abnormal storage activity and that important business files become inaccessible, leading the incident-response team to suspect ransomware execution. The publicly available **RanSMAP (Ransomware Storage and Memory Access Patterns)** dataset will be used to mimic the original incident in a controlled forensic environment.

The RanSMAP dataset contains low-level storage and memory access artifacts collected from ransomware and benign applications, including multiple **Conti ransomware variants**, as well as scenarios involving simultaneous execution of ransomware and legitimate applications. The dataset provides timestamped storage and memory access information that can be analyzed to reconstruct the behavioral sequence of a suspected ransomware event.

Python, Pandas, Jupyter Notebook, Matplotlib, and Seaborn will be used as the primary analysis tools. Python and Pandas will be used to process and correlate the forensic CSV artifacts, while visualization tools will be used to reconstruct temporal storage and memory activity. The purpose of these tools is to identify abnormal access patterns, compare ransomware behavior with benign application activity, examine high-entropy storage writes associated with potential encryption activity, and establish a behavioral timeline of the suspected attack.

The investigation will reconstruct the original ransomware scenario by selecting appropriate Conti-related artifacts from the RanSMAP dataset and analyzing them as evidence obtained from a compromised workstation. The study will correlate memory and storage access patterns to identify forensic indicators associated with Conti ransomware and reconstruct the sequence of malicious activity. The expected outcome is a reproducible forensic methodology for analyzing ransomware incidents using low-level memory and storage artifacts without executing live ransomware on a production or personal system.

Dataset: RanSMAP - Ransomware Storage and Memory Access Patterns

Primary ransomware: Conti

Dataset type: Storage and memory access-pattern CSV artifacts

Environment represented: Windows 10 controlled environment

Dataset scale: 11,820 CSV files representing 1,970 executions

Primary evidence: Storage read/write and memory read/write/execute patterns

Research paper: Hirano et al., *RanSMAP: Open dataset of Ransomware Storage and Memory Access Patterns*, Computers & Security, 2025.

Link : <https://github.com/manabu-hirano/RanSMAP>